Inverse of High Dimension Matrix

Recall

$$\underline{W} = (X^T X + \lambda I)^{-1} X^T \underline{Y}$$

if Feature vectors are of dimension d $\Rightarrow$ ($\underline{X}^T X$) is of size d+1 x d+1

The standard matrix inversion algorithm has a time complexity of $O(n^3)$, which means that the computation time increases rapidly with the size of the matrix (increasing cubically with the number of components).

For Example: if Features vector is of size 1000 then Inverse calculation is or order 1000,000,000 which is impractical

Solution:

*Instead of using **Least Squares** with Matrix inversion,*
*use **Gradient Descent** iterative solution*